

TalentVerify

Project Team

Muhammad Mobeen	21I-0444
Huzaifah Bin Saeed	21I-0432
Ahmed Raza	19I-0710

Session 2021-2025

Supervised by

Ms. Saira Qamar



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

June, 2025

Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Web Application	2
1.3.2	Emotion Detection Pipelines	2
1.3.3	Cheating Detection	3
1.3.4	Question Generation	3
1.3.5	Answer Evaluation	3
1.3.6	Avatar Generation	3
1.3.7	Dashboard	4
1.4	User Classes and Characteristics	4
2	Project Requirements	7
2.1	Event Response Table	7
2.2	Functional Requirements	8
2.2.1	Web application	8
2.2.2	Emotion Detection Pipeline	9
2.2.3	Cheating Detection	9
2.2.4	Question Generation	9
2.2.5	Answer Evaluation	9
2.2.6	Avatar Generation	10
2.2.7	Dashboard	10
2.3	Non-Functional Requirements	10
2.3.1	Reliability	10
2.3.2	Usability	10
2.3.3	Performance	11
2.3.4	Security	11
2.4	Domain Model	12
3	System Overview	13
3.1	Architectural Design	13
3.1.1	System Architecture	13

3.2	Design Models	15
3.2.1	Activity Diagram Diagram	15
3.2.2	Class Diagram	15
3.2.3	Class-Level Sequence Diagram	17
3.2.3.1	Employer Sets up Interview	17
3.2.3.2	Candidate Uploads Resume	17
3.2.3.3	System Analyzes Resume	18
3.2.3.4	Conduct Interview with Real-Time Emotion Detection	18
3.2.3.5	Cheating Detection During Interview	19
3.2.3.6	System Generates Interview Summary	19
3.2.3.7	Employer Reviews Interview Results	20
3.3	Data Design	20

List of Figures

List of Tables

Chapter 1

Introduction

This report lists the problem statement, scope, module divisions, requirements and system design for the first iteration of the project TalentVerify. TalentVerify is an AI interviewing web application that has a focus on integrity and evaluating soft skill score of candidates for well informed decision making. Providing an interface for both employer to setup the interview and the user to participate in the interview process, we hope to streamline the hiring process. TalentVerify will serve as an automated screening layer to filter out candidates in bulk and limit only qualified candidates for next stage for companies hiring process.

Utilizing the power of existing facial detection technology alongside audio embedding coupled with our own classification models we hope to bring our vision into life. TalentVerify will be able to evaluate soft skills in real-time while constantly verifying the integrity in the interview process by flagging suspicious activities.

1.1 Problem Statement

Conducting interviews for job positions can be time consuming and costly for any company looking to fill positions with quality candidates. This process has both an actual cost and an indirect cost that can be associated with it, Any company looking for the best possible candidate inevitably desires to interview as many good candidates as is feasible for them. The direct cost of conducting an interview is estimated to be anywhere from 70 to 200 dollars per candidate. The indirect cost may come in the form of lost time on understaffed projects and other such situations. This cost can build up to a very large amount, when any decent number of candidates is interviews. Existing systems on the market that aim to tackle this problem do not account for soft skills of a candidate and are similar to a question answer analysis bot. Our goal is to create is an AI agent to conduct interviews that can analyse behaviour, soft skills and provide a human face for the interviewee to

interact with.

1.2 Scope

TalentVerify is a web based application for employers to streamline the interview process. The scope of this iteration focuses on the creating and usage of deep learning models that can accurately identify audio and video emotions which will later be used for evaluating the soft skills.

An interface for both employer and regular user allows them to interact with the system. The employer sets the preferences to create an instance of the interview to begin the automate hiring process. Users will be authenticated and then allowed to partake in the interview.

During the interview process key interviewee features are extracted and evaluated using distinct models for comprehensive analysis of the candidates emotions. Moreover, a multiple entity detection system is also employed to find if the candidate is getting any external assistance.

1.3 Modules

1.3.1 Web Application

An interface that participants will use to interact with the system, consisting of a landing page that allows the user to either log in or create a new account. After the user is authenticated they are redirected to their relative interfaces, corresponding to their account type.

1. User Authentication
2. Employer Interface
3. Interviewee Interface

1.3.2 Emotion Detection Pipelines

The emotion detection pipeline consistent of creating and using existing deep learning models to evaluate emotions. The key frames and audio samples are transferred to the server during the interview process. These raw data are cleaned and transformed to extract relevant features which will be used make accurate classifications at key time frames.

1. Audio Emotions
2. Video Emotions

1.3.3 Cheating Detection

The same key frames from video emotion pipeline are used to check for one type of cheating method. This method involves having external assistance during the interview process by having another person present in the room when prompted to be in an isolated environment during the interview process.

1. Multiple entity detection
2. Deepfake detection

1.3.4 Question Generation

The uniqueness of the application is defined by how every interviewee and employer gets a tailored experience with respect to their resume and job description that is provided prior to the interview.

1. Resume Parsing
2. Job Description Parsing

1.3.5 Answer Evaluation

This module covers the evaluation aspect of the questions generated from the module above. The final step involved in creating a marking criteria to evaluate a candidate during the interview process.

1. Technical Evaluation
2. Non Technical Evaluation

1.3.6 Avatar Generation

Avatar Generation module will focus on creating an AI assistant that interviewees can interact with during the interview process. It will allow us to add a personal touch and replicate the environment of an in person interview

1. Avatar Video Generation
2. Avatar Audio Generation

1.3.7 Dashboard

The Dashboard module will be the final module for our end to end application. It will be a detailed dashboard that provides the employer of every candidates performance and the shortlisted candidates for the next stage of the hiring process. The amount of shortlisted candidates will be preset by the employer when they set the preferences while providing the job description.

1. Interviewees performance dashboard
2. Shortlisting feedback

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

The system will have 3 types of users:

1. Employer
2. Interviewee
3. Admin

User class	Description
Employer	An Employer is a companies' full time employee responsible for setting up the interview. In the real-world these people can be the HR, Technical recruiters or even tech lead themselves for smaller sized companies. Employer will only impact the system by starting the automated hiring process where they provide short listing criteria. Every organization can have either one or many Employers that can instantiate the hiring process.
Interviewee	After the Employer sets up the interview, the Interviewee can navigate the scheduled interview. The Interviewee is prompted to be in an isolated environment while also turning their webcam and microphone on. They participate in the interview by providing their resume and the interviewer agent asks and evaluates technical and non technical questions
Admin	System admin can overview the system by checking database logs and scheduled interviews. They can also access the hiring criteria set by the employers as well as monitor the shortlisting after the interview stage ends.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the TalentVerify.

2.1 Event Response Table

The Event Response table describes the functionality of the web application.

Event	Trigger	Response	Notes
Users Sign up	User submits valid credentials to register	Authenticate the given email and add information to the database	User specifies whether they are employer or interviewee
Users Logs In	User submits valid credentials to access the interface	Authenticate credentials and respond with user type	Different interface for employer and interviewee
Employer Sets Up Interview	Employer creates a new interview setup, specifying criteria	Save interview setup, notify interviewee with details	criteria include shortlisting number, interview domain and job description
Interviewee setup	Interviewee provides the resume	System responds that resume successfully received	The resume has to be in PDF format

Analyze Resume	Resume received from interviewee	Analyze the information to create a set number of technical and behavioral questions from resume and job description	The technical questions will be generated from job description and domain.
Real-time emotion detection	Interviewee is responding to the questions	Analyze the emotions from audio and video cues	The evaluation continues throughout the interview
Cheating Detection Trigger	Suspicious behavior detected during the interview	Trigger cheating detection, flag key frames and increment counter	Once the counter reaches a threshold value, the interview is marked as invalid due to cheating
Interviewee Completes Interview	Interviewee finishes answering all questions	Store interview results, process final emotion-based evaluation	Notify employer about interview completion
Employer Reviews Interview Results	Employer accesses the interview dashboard	Display interview summary, including emotion analysis and soft skill evaluation	Includes video playback and emotional trend analysis

2.2 Functional Requirements

This section describes the functional requirements of the TalentVerify. Functional requirements will help understand how the different modules affect the stakeholders in our online interviewing platform.

2.2.1 Web application

Following are the requirements:

1. The interviewee shall be able to view all interviews created by the employer.
2. The Employer shall be able to set up and interview by setting filtering criteria.

3. The system shall authenticate users' credentials and provide appropriate interfaces based on their role.
4. The Interviewee shall be able to filter Interviews by name and organization.

2.2.2 Emotion Detection Pipeline

Following are the requirements:

1. The system shall be able to identify emotions from video key frames.
2. The system shall be able to identify emotions from audio samples.

2.2.3 Cheating Detection

Following are the requirements:

1. The system shall be able to detect multiple facial entities during the interview process.
2. The system shall be able to detect deepfake faces from a given video key frame.

2.2.4 Question Generation

Following are the requirements:

1. The Interviewee shall be able to receive personalized behavioral questions based on their resume.
2. The Employer should be able to generate technical questions based on the job description.

2.2.5 Answer Evaluation

Following are the requirements:

1. The system shall be able to verify the facts in a response of technical question.
2. The system shall be able to score answers to behavioral questions.

2.2.6 Avatar Generation

Following are the requirements:

1. The interviewee shall be able to view an avatar during the interview.
2. The interviewee shall be able to listen to the avatar during the interview.

2.2.7 Dashboard

Following are the requirements:

1. The Employer shall be able to view candidates performance after the interview.
2. The Employer shall be able to view the shortlisted candidates after the interview.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality affect the overall performance of

2.3.1 Reliability

The system needs to maintain a non interrupted uptime for a smooth user experience:

REL-1: The application shall have a mean time between failures (MTBF) of at least 100 hours under average traffic conditions.

REL-2: In the event of a failure, the system shall recover and resume normal operations from the last checkpoint within 3 minutes, the interview process will need to restart instead of resuming.

REL-3: System failures during an interview shall trigger an automatic error log, notifying the admin while reverting back to original checkpoint before interview starting stage.

2.3.2 Usability

The application should be user-friendly for both employers and interviewees with minimal learning curves.

USE-1: The system shall allow first-time users to successfully navigate through the interview setup and participation process without needing external documentation or help,

achieving a first-time completion rate of 90%.

USE-2: The employer's dashboard for reviewing interview results shall be accessible and understandable within 2 minutes of use for a new user.

USE-3: The system shall display clear error messages in the event of invalid inputs and offer suggestions for correction, reducing input errors by 80%.

2.3.3 Performance

The system should perform efficiently under load, especially during concurrent interview sessions and data processing.

PER-1: 95% of interview sessions shall have less than 2 second of delay for real-time emotion detection feedback during video and audio processing.

PER-2: The resume analysis function shall generate interview questions based on the resume and job description within 10 seconds of submission.

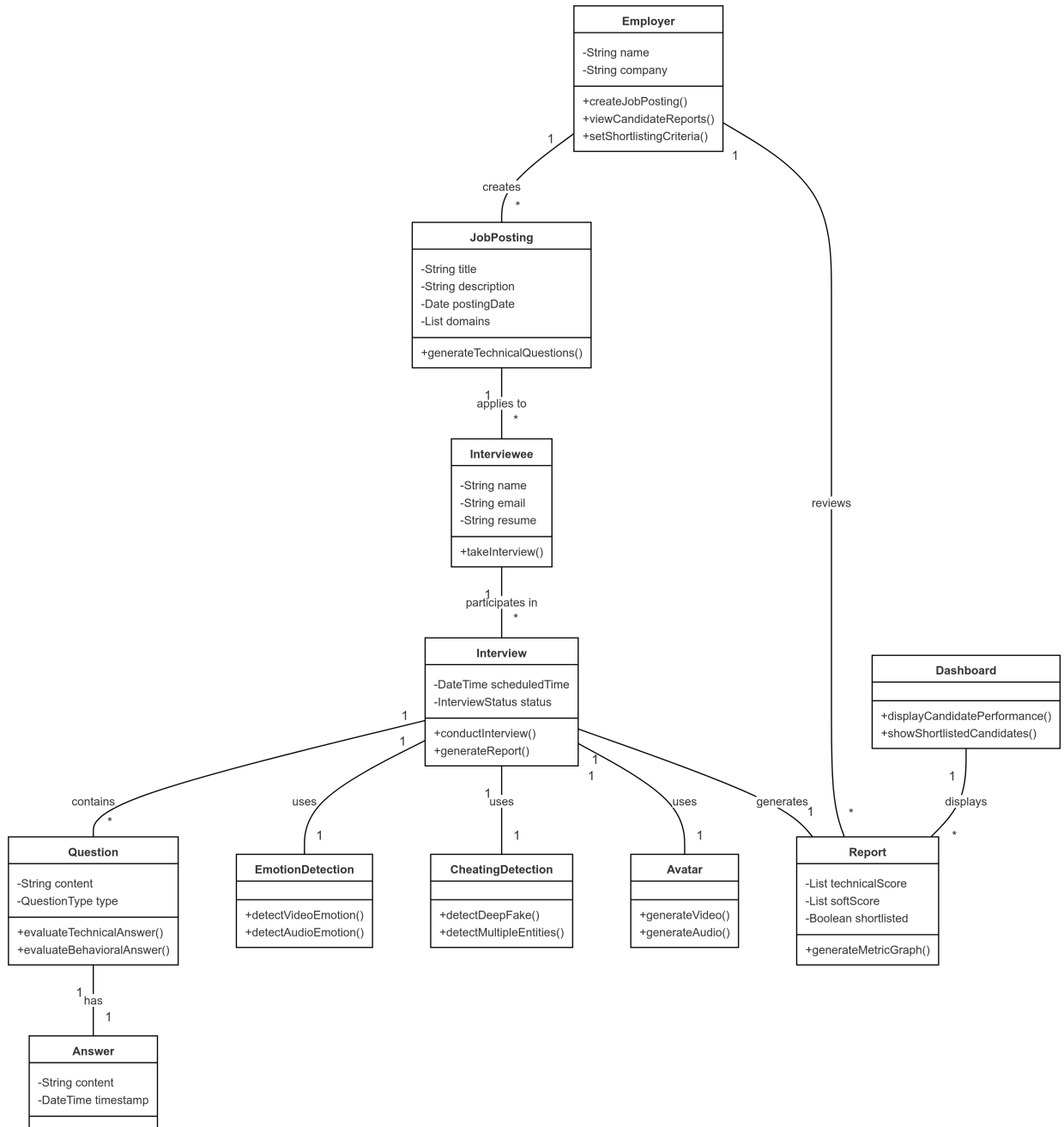
2.3.4 Security

The application should protect sensitive data, especially personal and interview-related information, ensuring the highest levels of confidentiality and integrity.

SEC-1: The application shall have role-based access control, ensuring that only authorized users (employers, interviewees, and system admins) can access specific data and functionalities.

SEC-2: Sensitive information such as passwords shall be encrypted and stored in the database.

2.4 Domain Model



Chapter 3

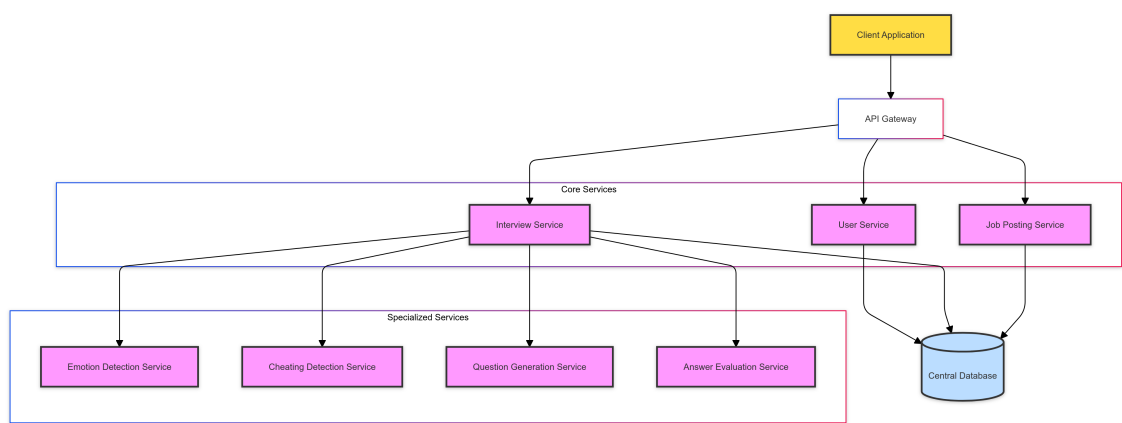
System Overview

Our system allows employers to conduct easy automated interviews. The employer can choose how many candidates to pass on, as well as the soft and technical skill score thresholds. The employer may invite candidates to participate in these interviews by linking them on the job posting. Our web app will then setup and conduct automated interviews of all the candidates and screen out the number specified by the employer based on how well they did on both their technical skills and their soft skills(confidence, nervousness, clarity etc). In the course of holding each interview, our automated agent will parse through each candidates resume, and generate personalized questions to assess each candidate as well as questions based on the job description provided by the employer. These candidates are then provided to the employer alongside data on how the interviewees in general performed on both their soft and technical skills.

3.1 Architectural Design

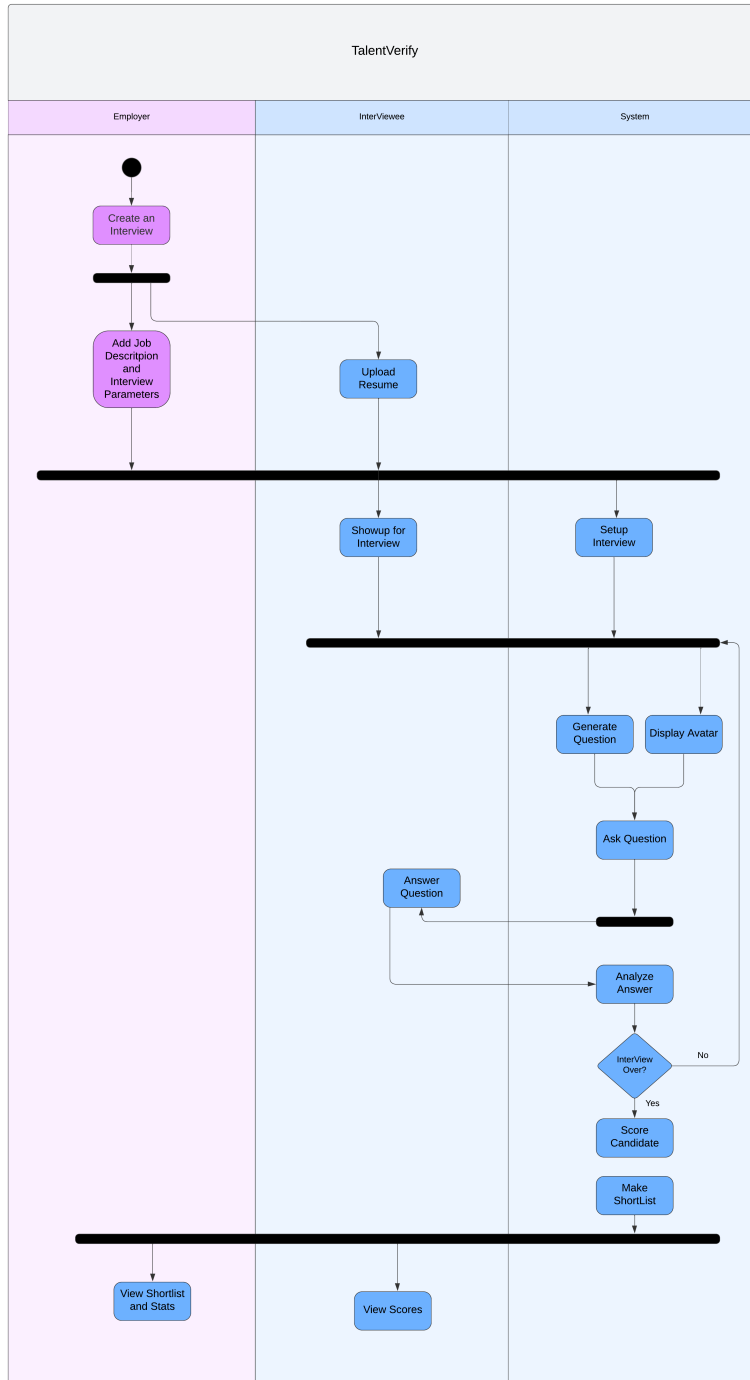
We chose a micro-service architecture for our application, with 3 core services, Job Posting, User, and Interview Conduction. Interview conduction also contains some critical sub-services that all need to work for it to function. We chose this architecture because it is important in our case to ensure that independent tasks can be completed even if some functionality of the system is unavailable. For example, an employer should still be able to view their dashboard and candidate shortlists, even if the job posting service is unavailable. Micro-service architectures also have the added benefit of scaling well, and dealing with asymmetrical load on different parts of the application.

3.1.1 System Architecture



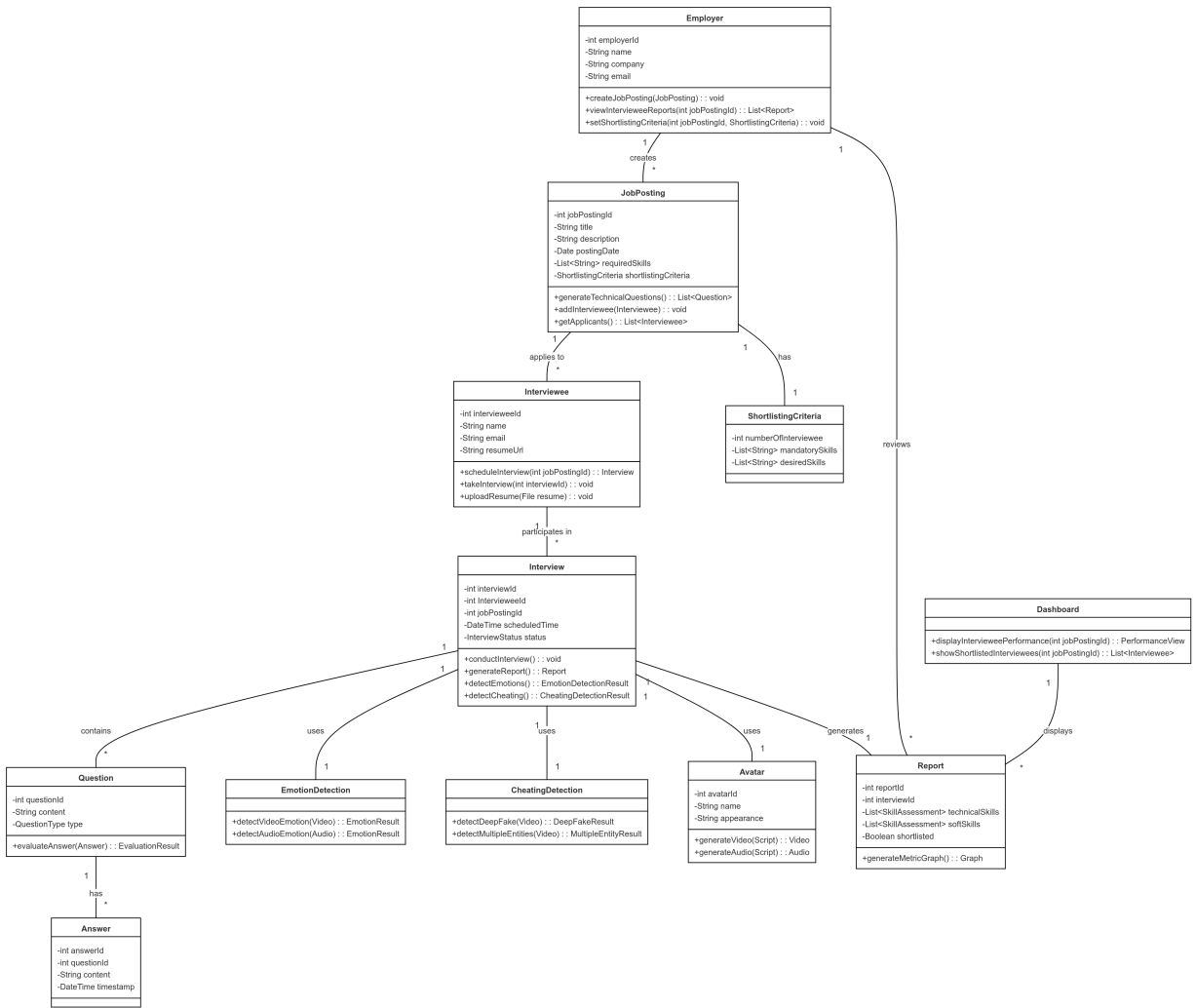
3.2 Design Models

3.2.1 Activity Diagram Diagram



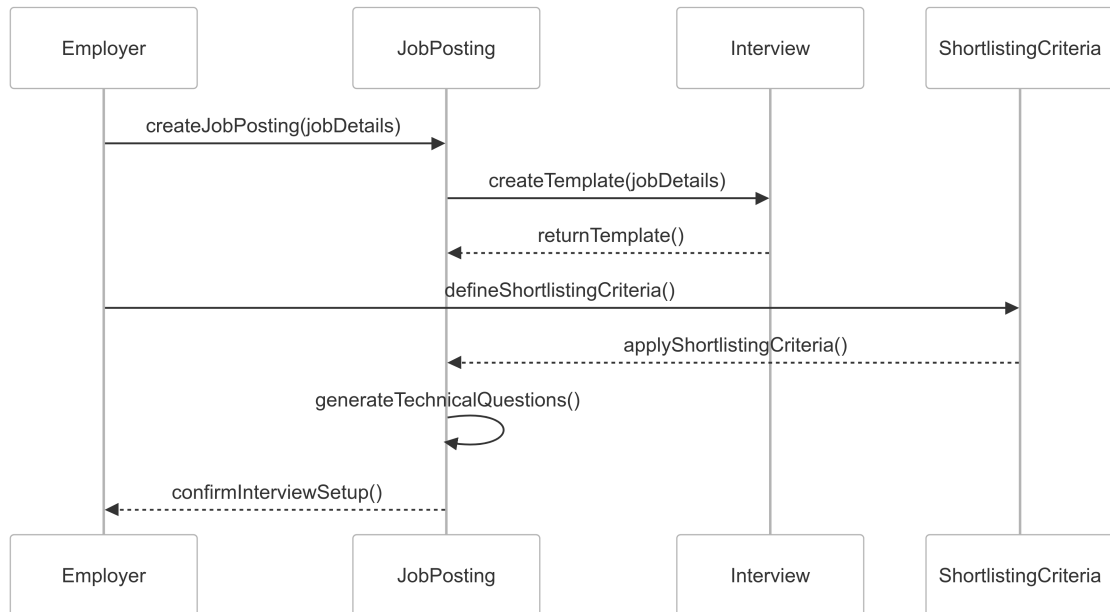
3.2.2 Class Diagram

3. System Overview

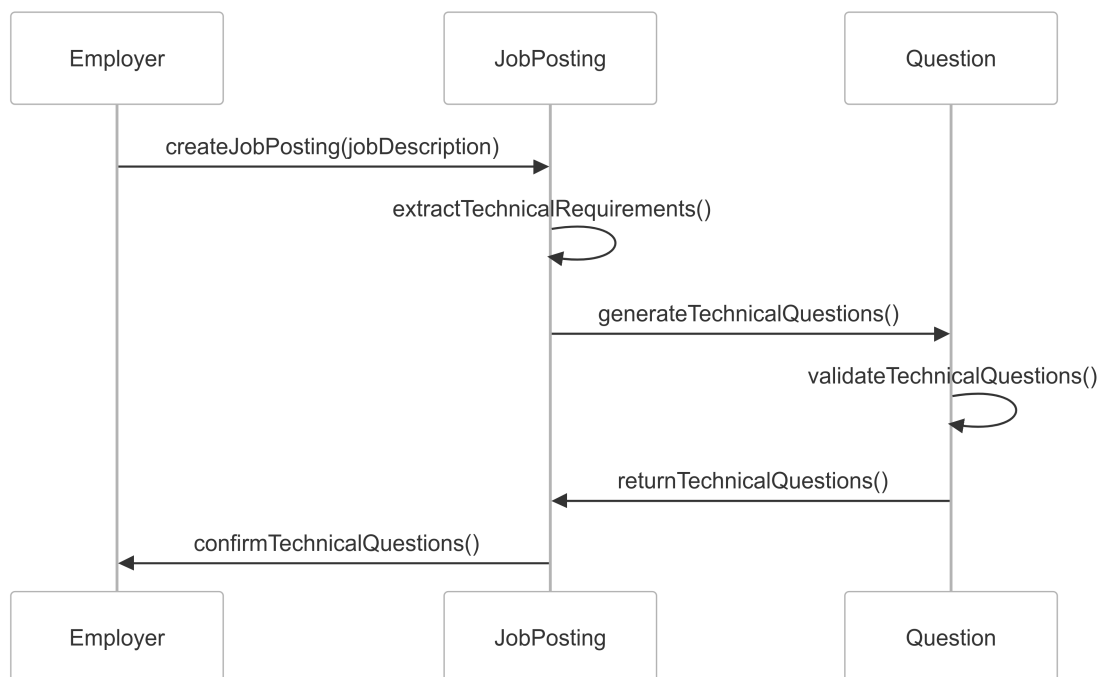


3.2.3 Class-Level Sequence Diagram

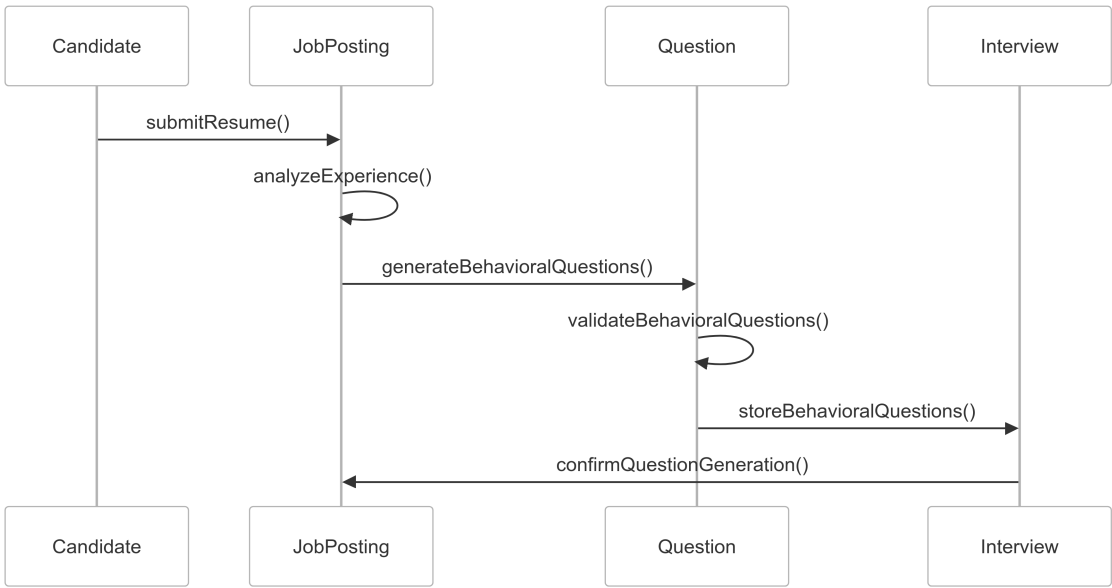
3.2.3.1 Employer Sets up Interview



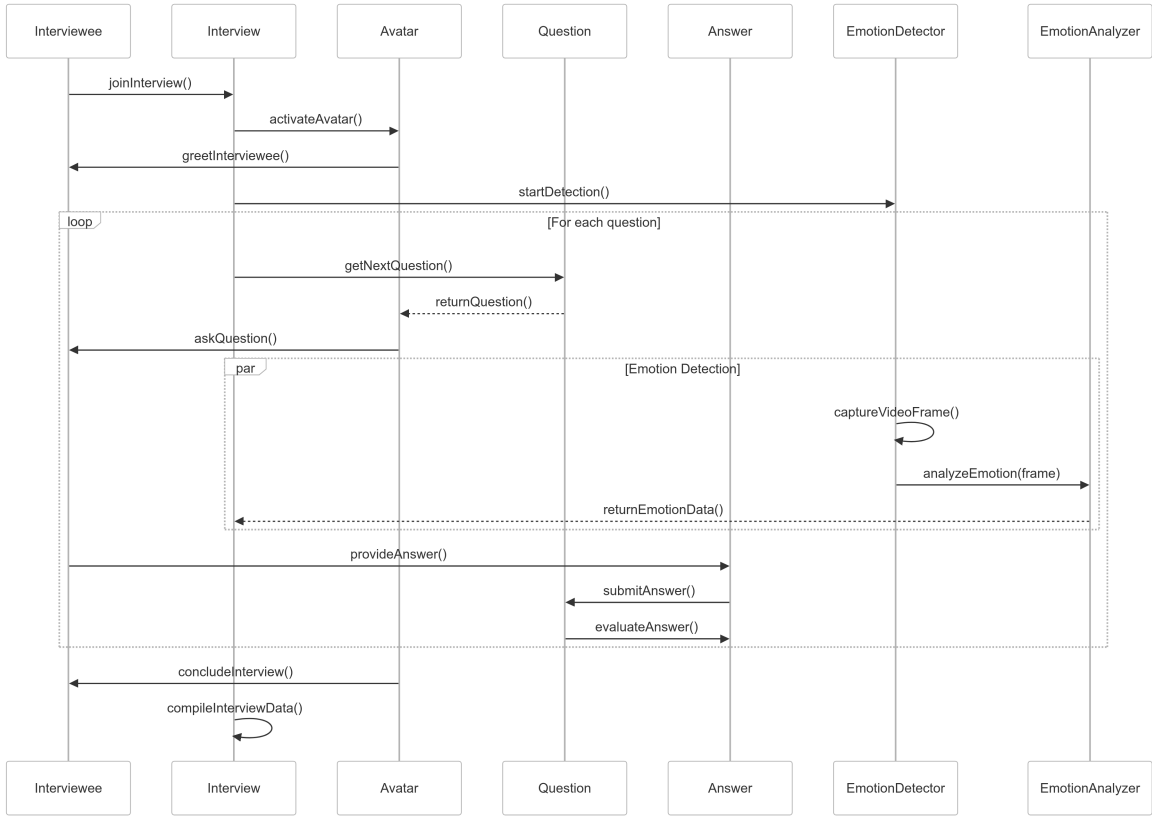
3.2.3.2 Candidate Uploads Resume



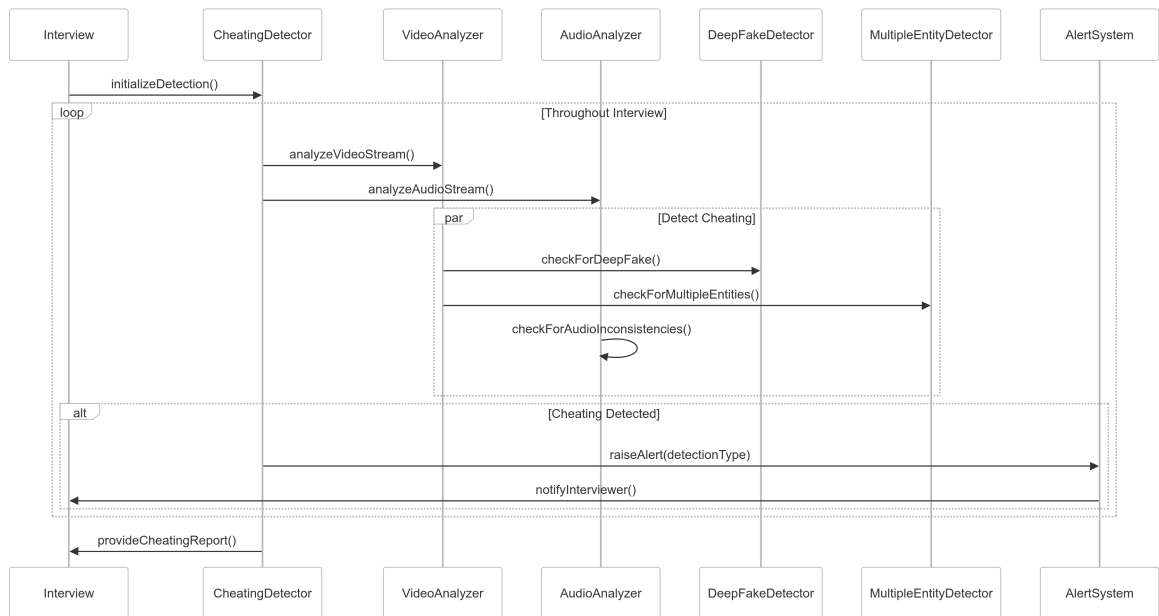
3.2.3.3 System Analyzes Resume



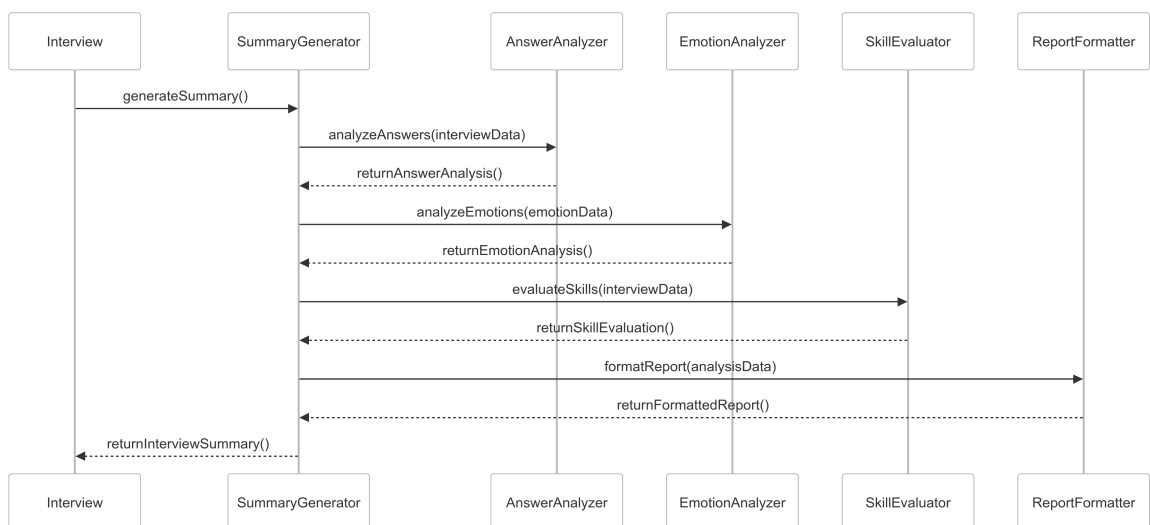
3.2.3.4 Conduct Interview with Real-Time Emotion Detection



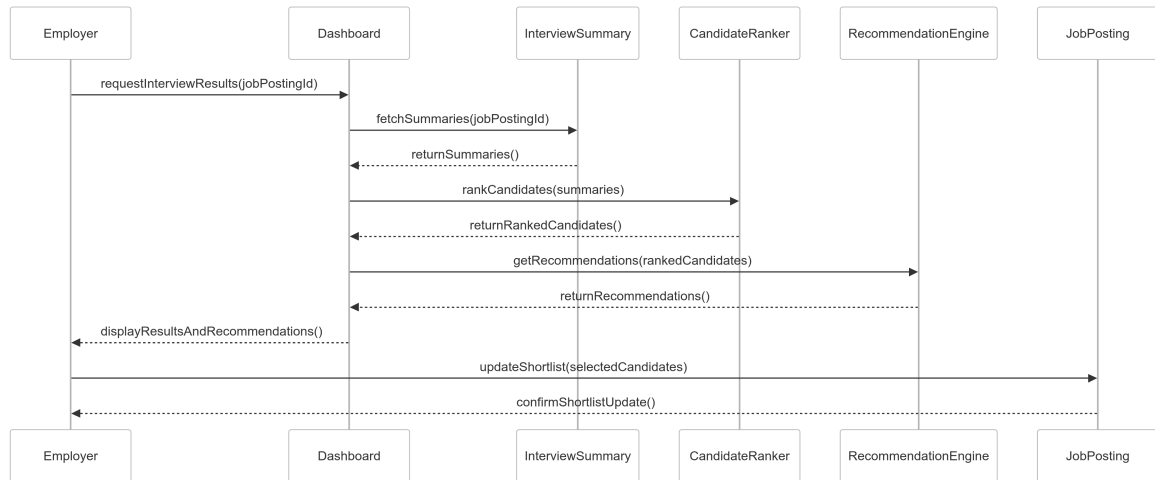
3.2.3.5 Cheating Detection During Interview



3.2.3.6 System Generates Interview Summary



3.2.3.7 Employer Reviews Interview Results



3.3 Data Design

Our database management system of choice is MongoDB. We chose MongoDB as it is a great free cloud based service with great scaling support. It is also NoSQL which allows us to store things like avatar parameters and weights easily. All of our user data is stored on a central database. Our database contains all user information, past and present scores, past and present interviews, statistics on candidate performance. All weights and necessary parameters for interviewer avatars are also stored inside of this database. Whenever an employer sets up an interview, we store this request along with a processed version of the job description. This processed version is generated by passing the description through an LLM to extract only necessary information. The same processing is applied to uploaded resumes, which are stored alongside their generated scores and any remarks generated by our scoring system.

All of our neural networks are stored on servers in the form of weight files, with backup copies on offline systems. All temporary data is stored server-side whenever necessary.